



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : MEL G631
Course Title : Physics and Modeling of Microelectronic Devices
Instructor-in-Charge : Rahul Kumar (rahul.kumar@pilani.bits-pilani.ac.in)
Instructor(s) : NA
Tutorial/Practical Instructors: Rejaul Ali, Khichar Anita Omprakash

1. Course Description: Physics and properties of semiconductor - a review; Semiconductor transport model, modeling of metal-semiconductor junctions, pn homojunction and heterojunctions, bipolar transistor, JFET, MESFET, HEMT and; MOS capacitor, modeling of MOSFET, scaling issues, and recent trends in advanced MOSFETs

Laboratory Components: Modeling and simulation of silicon-based devices with TCAD software. I-V, C-V measurement of MS and MOS junctions.

2. Scope and Objective of the Course:

This course is designed to provide the basic physical concepts behind microelectronic devices and imparts information on modeling of these devices for their use as circuit elements in integrated circuits. It will provide the ability to deal with advanced concepts of microelectronic devices along with non-ideal effects in devices with an attempt to establish an introduction to research methodology.

3. Text Books:

TB: R. S. Muller and T. I. Kamins, Device Electronics for Integrated Circuits, 3rd ed. Wiley, New York

4. Reference Books:

R1: Chenming Calvin Hu, "Modern Semiconductor Devices for Integrated Circuits", Pearson Education.

R2: Donald A. Neaman and D. Biswas, "Semiconductor Physics and Devices", 4th ed., Tata Mc Graw Hill

5. Course Plan:

Modules	Lec. No.	Topics to be Covered	References	Learning Outcome
Semiconductor Electronics	1-2	Basics of Semiconductors, Band and Bond model of solids	TB: 1.1 R1: 1.2-1.5	Students should identify the key parameters that affect the conduction & transport of carriers in semiconductors
	3	Thermal-equilibrium statistics, photogeneration & Quasi-Fermi levels	TB: 1.1 R1: 1.6-1.10	
	4-6	Free Carriers in semiconductors, scattering, velocity saturation	TB: 1.2 R1: Ch-2	



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Metal-Semiconductor Contacts	7-8	Equilibrium in electronic systems, idealized metal semiconductor junctions: Band diagram.	TB: 3.1, 3.2 R1: Ch-4 (part-III)	Understanding the growing importance of the design of high-performance transistors
	9-10	C-V Characteristics: Schottky contacts and ohmic contacts.	TB: 3.3, 3.4	
	11	Surface Effects on semiconductor contacts, depletion layer model	TB: 3.5	
pn Junctions and junction devices	12-13	Graded impurity distributions, Quasi-Neutrality Approximation; depletion layer model	TB: 4.1, 4.2 R1: 4.1, 4.2	To understand the concepts of the depletion region and minority carrier injection
	14-16	Heterojunction; p-n junction under bias; junction breakdown	TB: 4.3 & 4.4	
	17	p-n junction Devices: JFET & MESFET	TB: 4.5 R1: 4.3-4.6	
Currents in p-n junctions	18-19	Continuity equation, generation and recombination: SHR; Auger; surface	TB: 5.1, 5.2 R1: 4.7-4.9	To know the concepts of generation and recombination in p-n junctions
	20-21	Current-Voltage Characteristics of pn junction	TB: 5.3	
	22	Current through heterojunctions	TB: 5.5	
Properties of the MOS system	23-24	MOS structure and analysis of the ideal MOS structure.	TB: 8.1, 8.2 R1: 5.1, 5.2	To understand the concept underlying the flat-band and threshold voltages of MOS devices
	25-26	MOS Electronics: Equilibrium & non-equilibrium analysis	TB: 8.3	
	27-29	MOS Capacitance; Non-ideal MOS system: Oxide charges in MOS, Charge-coupled devices (CCD)	TB: 8.4-8.5, 8.7 R1: 5.6-5.10	
MOS Transistor	30-31	Gradual channel approximation, body effect, MOSFET I-V Model, subthreshold conduction	TB: 9.1 R1: 6.6 R2: Ch-2	To understand the effects of scaling in MOSFETs and the emergence of novel devices.
	32	Improved models for short channel MOSFETs; Mobility degradation; velocity saturation	TB: 9.2	



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	33-35	Multigate MOSFETs: UTB MOSFETs; FinFET and MOSFET Compact Model for circuit Simulation	TB: Ch-10 R1: Ch-7	
Bipolar Transistor: Device Modeling	36-37	Operation; I-V characteristics; Gummel number, high-level injection effect, current gain, heavy doping effect	TB: 6.1, 6.2 R1: Ch-8	To understand the modeling of BJT and to know the integrated-charge model
	38-40	Ebers-Moll Model; Early effect & Charge control model	TB: 6.4, 7.1, 7.4 R1: 8.5, 8.6, 8.10	

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (CB/OB)
Mid-Semester Test	90 min.	20%	08/10/2026 FN2	CB
Lab and project Components - Regular lab performance - Research project - Final Lab quiz/Viva	Weekly	40%	To be announced in class	OB
Quizzes	10-15 min.	10%	Total 5	CB
Comprehensive exam.	120 min	30%	11/12/2026 AN	OB+CB

7. Chamber Consultation Hour: Every Saturday, 11-12 AM.

8. Details of the laboratory components: Separate details will be uploaded on Nalanda.

9. Notices: Notices for the course will be displayed on the EEE Notice Board and NALANDA.

10. Make-up Policy: Make-up will be allowed for genuine cases. Prior application should be sent for seeking the same.

11. Note (if any): NIL

Instructor-in-charge
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